PY-STEM · ACTIVITY PYB-03 · MECHANICS

Pulley Rescue Relay

Lift the load with smarter force, not stronger arms.

Win condition: Best lift with correct setup and explanation.

THE SCIENCE YOU ARE USING

Mechanical advantage. A single fixed pulley just changes the direction of your pull. Add movable pulleys and you trade distance for force: you pull more rope but with less effort. More supporting rope segments means more mechanical advantage.

Force and direction. Pulleys let you pull down to lift up, and let a small force raise a big load. You build pulley systems, compare the effort needed, and explain why the setup helps.

HOW TO BUILD AND RUN IT

01 **Lift with no pulley.** Measure the effort to lift the load by hand as a baseline.

02 **Add a fixed pulley.** Reroute over one fixed pulley. Note that direction changes but effort does not.

03 **Add a movable pulley.** Build a system with a movable pulley and feel the effort drop.

04 **Measure the advantage.** Use a spring scale to compare effort across setups. Keep the same load heavy enough that the scale reads clearly, not near zero.

05 **Explain the tradeoff.** Show how more rope segments cut the force you need.

Safety: Keep feet clear of loads. Allergy: None.

MATERIALS

Classroom pulleys	per team
String	shared
Weights	shared
Spring scales	if available

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________

The ONE variable we are testing:

Baseline result: ______________________

After redesign: ______________________

DEFEND YOUR DESIGN

What evidence made you change your design? ______________________

If you ran it again, what is the ONE thing you would change? ______________________

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Setup accuracy	25
Explanation	20
Teamwork	20
Total	**100**